

LD306S Radar Module Command Description V1.1

All commands are hexadecimal format commands for firmware version V1.0.5 and above.

Parameter Setting Command Format:

character threshold	header	address	command number	command parameter	checking	end of frame
byte length	2	1	1	2	1	2
content	BAAB	00	E5	XX	00	55BB

For ease of use, the command format is fixed byte in all fields except command number and command parameters. The corresponding command functions are listed in the table below:

Table 1 Common Setup Instructions

function	command number (1 byte)	command parameter (2 byte)	description
Minimum detection speed	E0	Value range: 0~200, unit: km/h, default 10km/h	设置模块检测的最小速度
Max detection speed	E5	Value range: 50~350, unit: cm, default is 350cm	设置模块检测的最大速度
(level of) sensitivity	E1	Value range: 0~20, unit : gear, default is 4 gears	设置模块检测灵敏度, 参数值越低, 检测距离越远, 0 为最远
communications protocol	F6	Take values 7 and 8. 7: hexadecimal (HEX) protocol 7: String (ASCII) protocol	The format of communication data can be modified by modifying the protocol type.
response time	D1	400msValue range: 400~3000 Unit: ms default 400ms	Setting the data reporting interval
latency	E4	Take values: 200~3000 Unit: ms Default 800ms	Set no target judgment delay time, continuous detection of 800ms all no vehicle found then speed to zero
Setting the ID address	DB	Value range 0~255	Setting the device ID address number

Table 2 Advanced Settings Command Table

Function	Command number (1 byte)	Command Parameters (2 bytes)	description
Inspection Direction	B1	Takes the value 0 or 1 0: close to the direction 1: away from the direction 2: Bi-directional detection Default detection of near direction	Setting the direction of travel of the detected target vehicle
fitting factor	B3	Take the value 1~65535, unit: 0.001 Default is 3000, i.e. the slope is 3	The distance calibration is calibrated using a single equation, and the fitting coefficient is the slope k
bias correction	B4	Take the value 0~65535, unit: 0.01km/h Default is 1957, i.e. 19.57km/h	The bias value in the calibration equation b
output level	D2	Value: 0 or 1 0: 0V with car 1: 5V with car	Setting the level signal for the presence or absence of the vehicle
baud	D4	Value	Setting the serial port communication baud rate
		48	
		96	
		192	
		384	
		576	
		1152	
output target	D9	Value: 0 or 1 or 2 0: nearest vehicle 1: fastest vehicle 2: slowest vehicle	Setting the target vehicle for the output result
restore the default	DE	Value 0	All parameters restored to factory default settings
Number of extreme statistics	E7	Values: 0~100 Unit: times Default: 0 times	Number of measurement statistics for extreme value processing
Number of Extreme Filtering	E8	Value: less than the number of counts Unit: times Default: 0 times	Total number of times the maximum and minimum values are filtered out
Firmware Upgrade	DF	Value: 1	Firmware Upgrade
Query parameters	FE	Value: 0	Query all current parameters of the module

- Note: Command parameters need to be converted to hexadecimal (accounting for 2 bytes) format, such as setting the minimum detection distance to 50km/h.
Then the command is: BAAB 00 E0 0032 00 55BB

Set successfully return:

0D0A7772697465206F6B0D 0A that is, ASCII code "

enter + writeok + enter".

The result of the query parameter is 108 bytes in total, which is parsed in the following table:

Byte 0~1	Byte 2	Byte 3	Byte 4-5	Byte 6-7
header (FAFB)	address (00)	Parameter length (64)	Parameter header (FAFB)	operating mode
Bytes 8 to 9	10th to 11th bytes	Byte 12~13	Byte 14~15	Byte 16~17
baud	XXXX	fitting factor	bias correction	Firmware version number
Bytes 18 to 55	Bytes 56 to 57	Bytes 58 to 61	Bytes 62-63	Bytes 64 to 65
XX...XX	Inspection Direction	XX...XX	XXXX	XXXX
Bytes 66 to 67	Bytes 68 to 69	Bytes 70 to 71	Bytes 72 to 73	Bytes 74 to 75
XXXX	XXXX	XXXX	Number of extreme statistics	Number of Extreme Filtering
Bytes 76 to 77	Bytes 78 to 79	Bytes 80 to 81	Bytes 82 to 83	Bytes 84-85
Number of slides	Protocol type	XXXX	output target	response time
Bytes 86 to 87	Bytes 88 to 89	Bytes 90 to 91	Bytes 92 to 95	Bytes 96 to 97
(level of) sensitivity	XXXX	Delayed zero time	XXXX	Maximum detection speed
Bytes 98 to 99	100th to 101st bytes	Bytes 102 to 103	Bytes 104 to 105	Bytes 106 to 107
Minimum detection speed	output level	Parameter tail (00EF)	checksum	End of frame (AA55)

XXXX is not necessarily zero, the specific parameters are not open yet;

Byte 104~105 (sum check): sum of all bytes from byte 0 to byte 103, high bit first.

Attachment:

When the radar is used in protocol 6, the radar is in query mode and shall send a query command: 555A02D384 Where: 555A: fixed byte;

02: command number

D3: device ID address

84: the first 4 bytes and parity low 8 bits